Name: Sovan Chhorponn
ID: e20220873

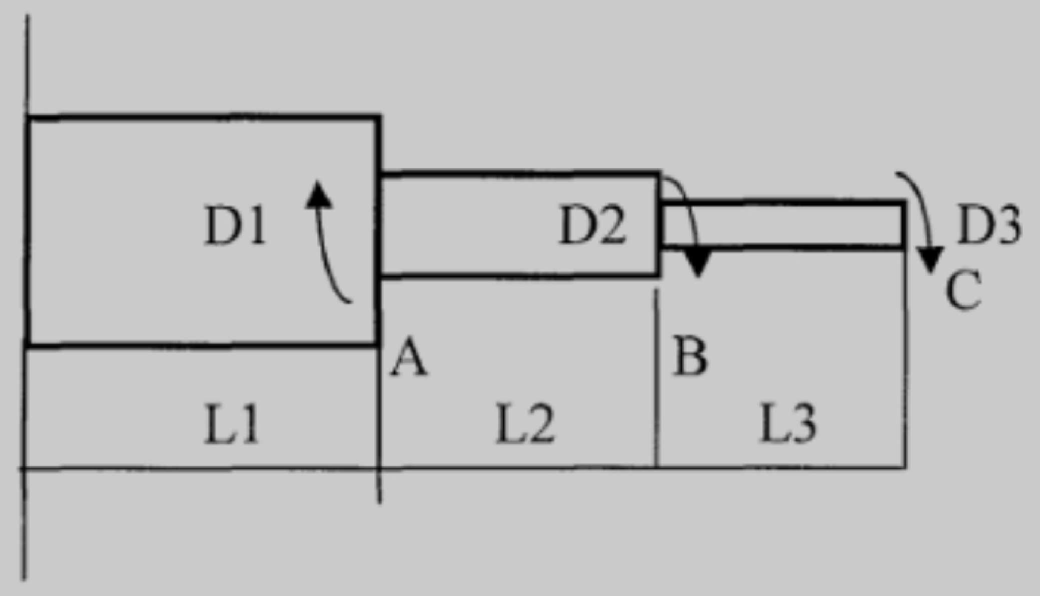
# Assessment

3. The cylindrical steel shaft shown in the picture is subjected to three different torsional moments.

a) Draw the torsional moment diagram and find the maximum shear stress
b) Find the relative rotation between point C and B.
c) Find the principal stresses for the critical section.

$T_1 = 30kN.m$
$T_2 = 15kN.m$
$T_3 = 10kN.m$
$d_1 = 30cm$
$d_2 = 25cm$
$d_3 = 15cm$

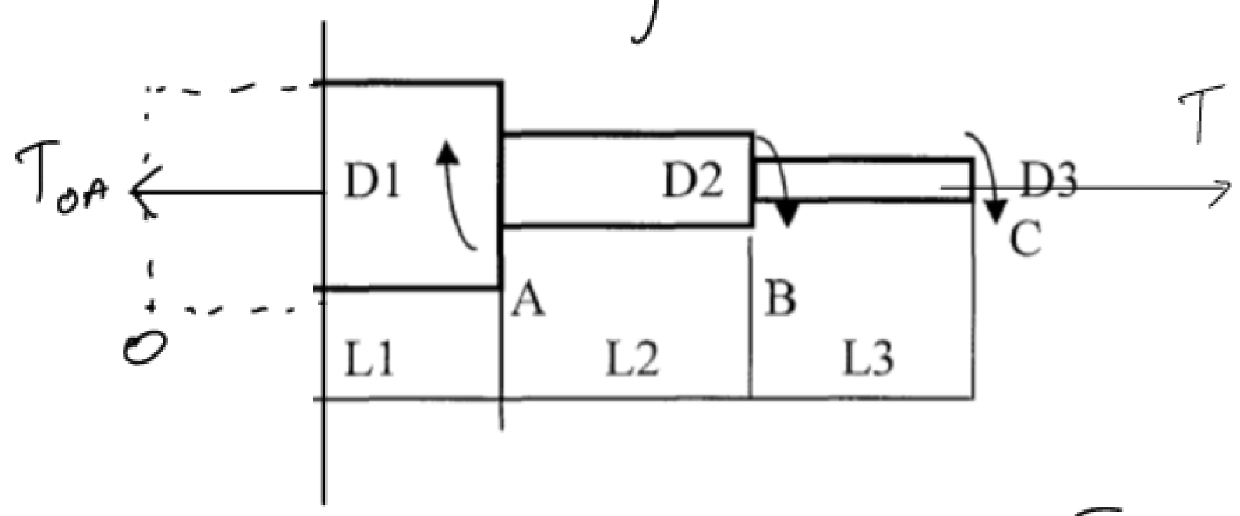
$L_1 = 1m$
$L_2 = 1.5m$
$L_3 = 1m$




## Answer

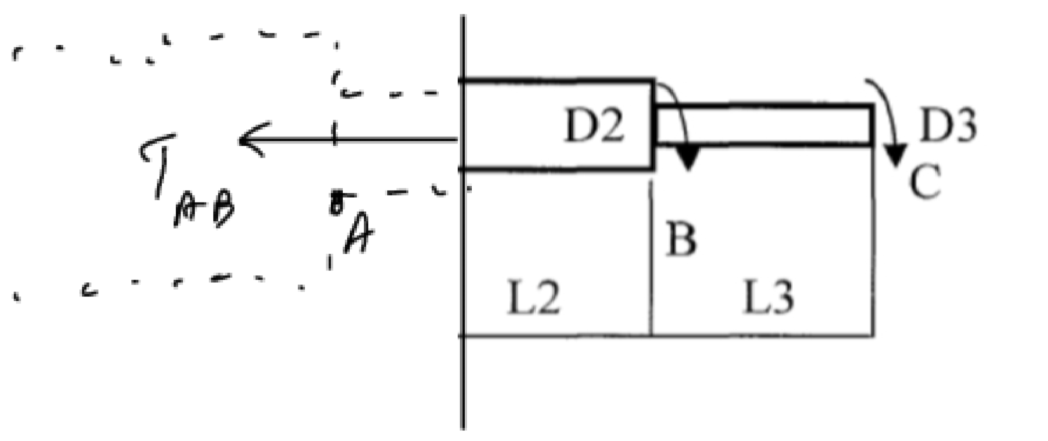
a). Draw torsional moment diagram and find maximum shear stress

+ Calculate for T in each region



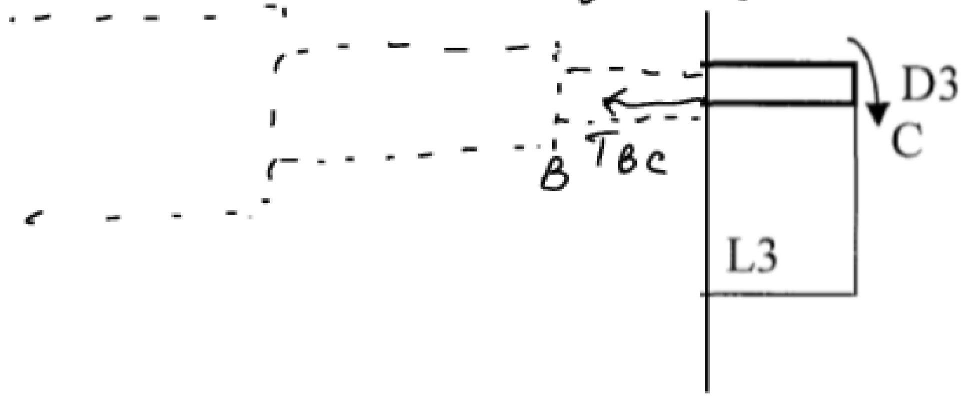

$0 = -T_{OA} - T_A + T_B + T_C$

$\Rightarrow T_{OA} = -T_A + T_B + T_C = -30 + 15 + 10 = -5 KN.m$




$0 = -T_{AB} + T_B + T_C$

$\rightarrow T_{AB} = T_B + T_C = 15 + 10 = 25 \ KN.m$

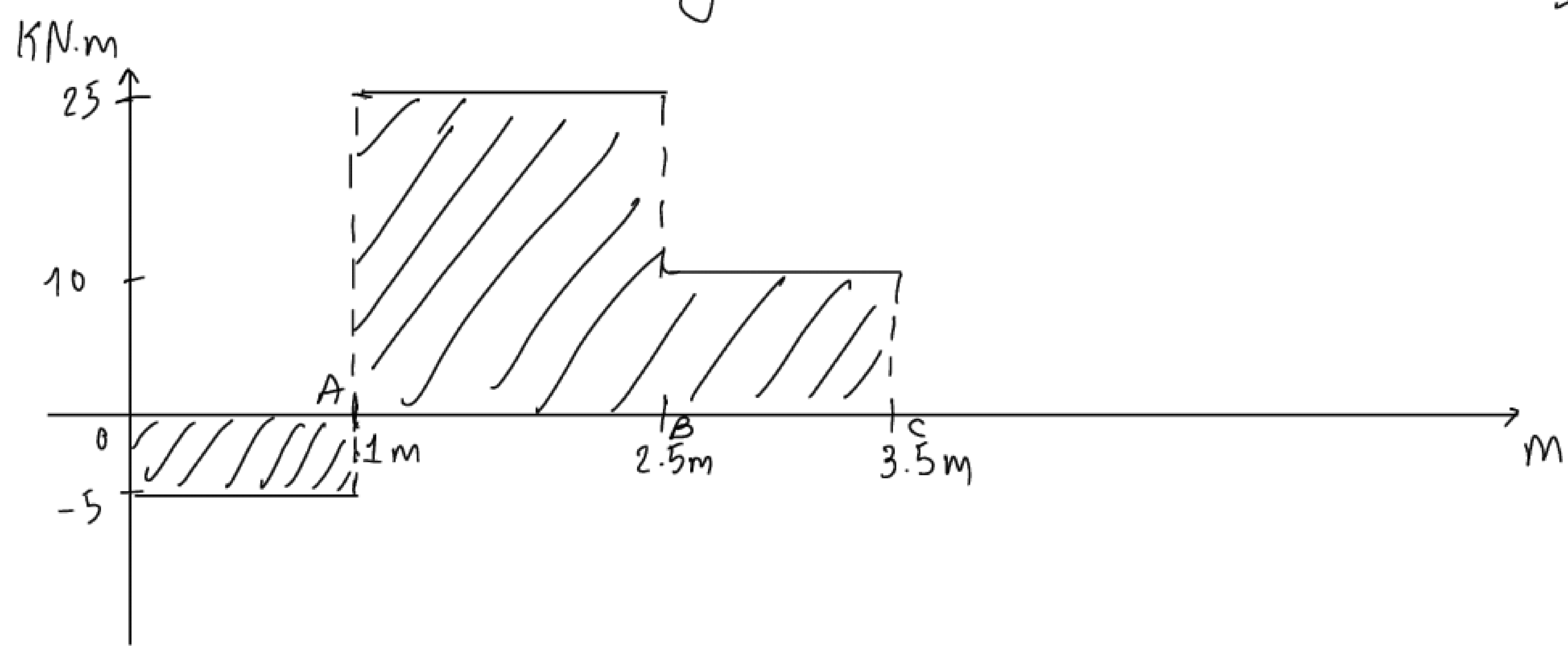



$0 = -T_{BC} + T_C$

$\Rightarrow T_{BC} = T_C = 10 \ KN.m$

## Torsional moment diagram

$$I_p = \frac{\pi D^4}{32}$$




+ find maximum shear stress

- $\tau_{OA} = \left|\frac{T_{OA} r_1}{I_p}\right| = \left|\frac{-5\times10^3 \times 15\times10^{-2}}{\frac{\pi \times (30\times10^{-2})^4}{32}}\right| = 943\,140.4035\, Pa$

- $\tau_{AB} = \left|\frac{T_{AB} r_2}{I_p}\right| = \left|\frac{25\times10^3 \times 12.5\times10^{-2}}{\frac{\pi \times (25\times10^{-2})^4}{32}}\right| = 8\,148\,733.086\, Pa$

- $\tau_{BC} = \left|\frac{T_{BC} r_3}{I_p}\right| = \left|\frac{10\times10^3 \times 7.5\times10^{-2}}{\frac{\pi \times (15\times10^{-2})^4}{32}}\right| = 15\,090\,246.460\, Pa$

$\Longrightarrow \tau_{max} = \tau_{BC} = 15.09 MPa$

b). Find relative rotation between C and B.

$$G = \frac{E}{2(1+\nu)}$$

$E = 210\, GPa$

$\nu = 0.3$

$\Rightarrow G = 8.0769\times10^{10}\, Pa$

Using formula: $\phi_{BC} = \frac{T_{BC} L_{BC}}{G I_p}$

$$= \frac{10\times10^3 \times 1}{8.0769\times10^{10} \times \frac{\pi}{32} \times (15\times10^{-2})^4}$$

$$= 2.491\times10^{-3}\, rad$$

$$= 0.1427^\circ$$

So, the relative rotation between C and B is $\boxed{0.1427^\circ}$

c). You said I can skip it.